

Structural Optimization Project

Finite Element Implementation and Multi-Algorithm Optimization

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Outline

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- 3 Validation
- 4 Conclusion

The Engineering Dilemma

Designers must balance two conflicting objectives:

- **Safety:** Structure must withstand loads (Stress & Stability).
- **Economy:** Structure must be lightweight (Material Cost).

General Definition

Find design variables x to:

- **Minimize:** $f(x)$
- **Subject to:** $g(x) \leq 0$

Methodology I: Finite Element Formulation

Element Stiffness Matrix

The structure is modeled using 2D bar elements. For an element e connecting nodes i and j at angle θ :

$$K^e = \frac{EA_e}{L_e} \begin{bmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{bmatrix}$$

where $c = \cos(\theta)$ and $s = \sin(\theta)$ transform local axial stiffness to global coordinates.

Global Assembly

The global stiffness matrix K is assembled by summing elemental contributions based on nodal connectivity.

Methodology II: Matrix Partitioning

1. Classification of Degrees of Freedom (DOFs)

To apply boundary conditions, we separate the global displacement vector u into two distinct sets:

- **Free DOFs (u_f):** Nodes that are free to move (Unknown displacements).
- **Constrained DOFs (u_c):** Supports with prescribed values (known displacements).

2. Partitioning the Equilibrium Equation

We rearrange the rows and columns of the global system $Ku = F$ to group these sets together:

$$\begin{bmatrix} K_{ff} & K_{fc} \\ K_{cf} & K_{cc} \end{bmatrix} \begin{bmatrix} u_f \\ u_c \end{bmatrix} = \begin{bmatrix} F_f \\ F_c \end{bmatrix}$$

Methodology III: System Resolution

3. Solving for Unknowns

Since the constrained displacements are zero ($u_c = 0$), the system simplifies significantly. We extract the top row of the partitioned matrix:

$$K_{ff}u_f + K_{fc}(0) = F_f$$

This yields the **Reduced Equation** for the free nodes:

Linear Solver Implementation

$$K_{ff}u_f = F_f \quad \implies \quad u_f = K_{ff}^{-1}F_f$$

Implementation Note:

In Python, we use `numpy.linalg.solve(K_ff, F_f)` which is computationally more efficient and stable than explicitly inverting the matrix.

Methodology IV: Optimization Problem Formulation

Objective Function

The primary goal is to minimize the total weight of the structure by adjusting the cross-sectional area of each bar.

Minimization Function

$$\underset{\mathbf{x}}{\text{minimize}} \quad M(\mathbf{x}) = \sum_{i=1}^m \rho L_i A_i$$

Design Variables ($\mathbf{x}$)

The vector $\mathbf{x} = [A_1, A_2, \dots, A_m]^T$ represents the cross-sectional areas of the m bars in the truss.

Design Space Bounds

Each bar is constrained by physical limits: $A_{\min} \leq A_i \leq A_{\max}$.

Methodology V: Constraint Implementation

Solvers (*SLSQP*, *COBYLA*, *COBYQA*) require constraints in the form $g(x) \geq 0$. We implemented the following constraints.

1. Yield Stress Constraint

Condition: $|\sigma_i| \leq \sigma_{yield}$

Solver Implementation:

$$g_1(x) = 1 - \frac{|\sigma_i(x)|}{\sigma_{yield}} \geq 0$$

2. Buckling Constraint

Condition: $|F_i| \leq P_{crit}$

Solver Implementation:

$$g_2(x) = 1 - \frac{|F_i(x)|}{P_{crit}(x)} \geq 0$$

Methodology VI: Optimization Scaling

Why Scale?

Cross-sectional areas ($A_i \approx 10^{-4} m^2$) are numerically very small. Gradient-based solvers (like SLSQP) struggle when variables and gradients differ by orders of magnitude from unity (1.0).

Scaling Strategy

We introduced a scaling factor A_{ref} to normalize the design space:

$$\tilde{x}_i = \frac{A_i}{A_{ref}} \quad \text{where } A_{ref} = 10^{-4} m^2$$

- **Input to Solver:** $\tilde{x}$ (Dimensionless values).
- **Inside Objective Function:** $A_i = \tilde{x}_i \cdot A_{ref}$

Result: Improved convergence speed and numerical stability of gradients.

Methodology VII: Stability Model

The Critical Constraint: Euler Buckling

For compressive members in the advanced problem, yield stress is insufficient. We implemented a non-linear buckling constraint assuming a solid circular cross-section ($I = A^2/4\pi$).

Critical Load Calculation:

$$P_{crit} = \frac{\pi^2 EI}{L^2} = \frac{\pi EA^2}{4L^2}$$

Validation: Setup, Parameters Assumptions

Parameters (Steel)

Param	Value
E	200 GPa
ν	0.266
ρ	7860 kg/m ³
σ_I	250 MPa
L	1 m

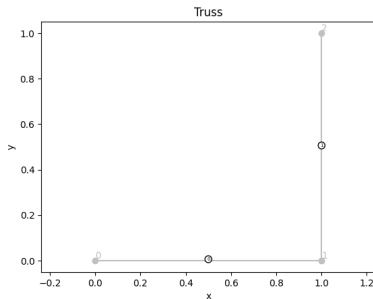


Figure: Initial Geometry

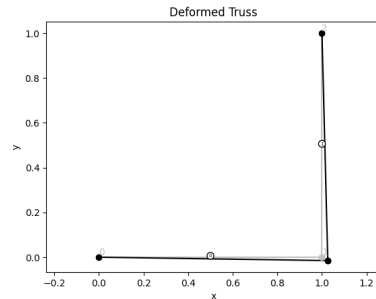


Figure: Deformed State

Validation: Optimization Results

Optimization Goal:

Minimize mass subject to **Yield Stress only** ($\sigma \leq 250$ MPa).

Converged Solution (COBYLA):

- **Mass:** 0.2558 kg
- **Status:** Fully Stressed Design (100% utilization).
- **Iterations:** 83

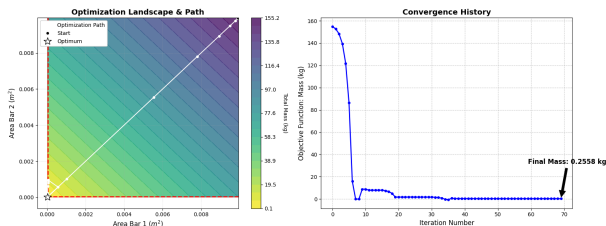


Figure: Convergence History

Project 2 Configuration

Setup (Data Table):

• External Load:

- Node 7
- $F_x = 0 \text{ N}$, $F_y = -6000 \text{ N}$

• Boundary Conditions:

- Fixed u_x : Nodes $\{0, 2, 3, 5, 6, 8\}$
- Fixed u_y : Nodes $\{0, 2\}$

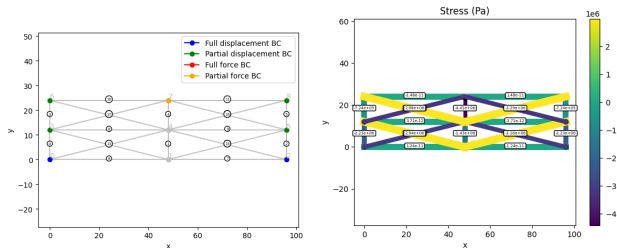
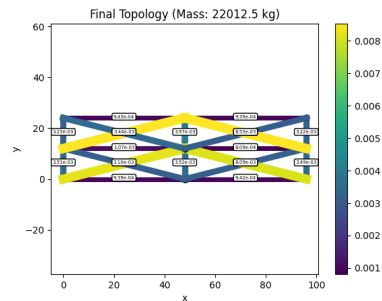
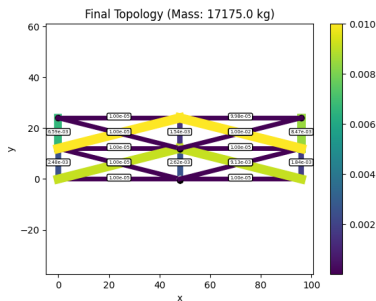
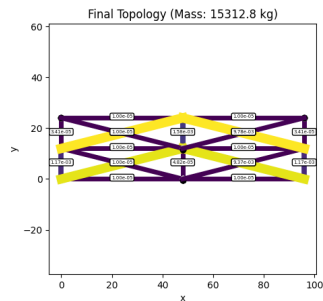


Figure: Setup & Initial Stress

Optimization Results: Algorithm Comparison



Verification (SLSQP)

Table: Critical Member Utilization (Selected Bars)

Bar ID	Area (cm^2)	Force (N)	Stress (%)	Buckling (%)	Status
0	11.69	-1,566	0.5	100.0	LIMIT
1	0.48	-3	0.0	100.0	LIMIT
2	11.69	-1,566	0.5	100.0	LIMIT
3	0.34	-1	0.0	100.0	LIMIT
4	15.84	-2,873	0.7	100.0	LIMIT
5	0.34	-1	0.0	100.0	LIMIT
12	93.67	-5,912	0.3	100.0	LIMIT
15	93.67	-5,912	0.3	100.0	LIMIT
16	97.82	-6,446	0.3	100.0	LIMIT
19	97.82	-6,446	0.3	100.0	LIMIT

Verification (COBYLA)

Table: Critical Member Utilization (Selected Bars)

Bar ID	Area (cm^2)	Force (N)	Stress (%)	Buckling (%)	Status
0	24.76	-1,637	0.3	23.3	OK
1	26.25	-3	0.0	0.0	OK
2	18.38	-1,637	0.4	42.3	OK
3	65.93	-1	0.0	0.0	OK
4	15.44	-2,732	0.7	100.0	LIMIT
6	0.10	-0	0.0	100.0	LIMIT
11	1.00	-1	0.0	100.0	LIMIT
12	91.33	-5,620	0.2	100.0	LIMIT
16	100.01	-6,738	0.3	100.0	LIMIT
19	100.00	-6,737	0.3	100.0	LIMIT

Verification (COBYQA)

ID	Area	Force	σ %	Buckl %	Stat
0	35.10	-1,950	0.2	13.8	OK
1	35.17	-747	0.1	5.3	OK
2	34.86	-1,912	0.2	13.7	OK
3	32.29	-413	0.1	3.5	OK
4	39.75	-3,672	0.4	20.3	OK
5	32.22	-375	0.0	3.2	OK
6	9.39	0	0.0	0.0	OK
7	9.42	-0	0.0	0.2	OK
8	10.73	-6	0.0	7.1	OK
9	8.09	4	0.0	0.0	OK

ID	Area	Force	σ %	Buckl %	Stat
10	9.43	-0	0.0	0.3	OK
11	9.39	0	0.0	0.0	OK
12	82.06	-4,480	0.2	98.7	OK
13	31.05	1,539	0.2	0.0	OK
14	31.05	1,540	0.2	0.0	OK
15	80.87	-4,332	0.2	98.3	OK
16	85.28	-4,799	0.2	97.9	OK
17	34.39	1,704	0.2	0.0	OK
18	30.50	1,546	0.2	0.0	OK
19	85.29	-4,799	0.2	97.9	OK

Conclusion

- ❶ **Methodology:** Developed a robust FE solver with matrix partitioning and non-linear stability constraints.
- ❷ **Physics: Euler Buckling** ($P \propto A^2$) is the driving design constraint, requiring significantly more mass than simple yield criteria.
- ❸ **Optimization:**
 - **SLSQP** proved superior for this smooth, constrained problem, achieving the global minimum (15,312 kg).
 - Derivative-free methods struggled to eliminate non-structural members.

Questions?